

Detailed Notes: Newton's Laws of Motion

A comprehensive study note covering Newton's three laws, inertia, force, momentum, free-body diagrams, friction, tension, circular motion, impulse, conservation of momentum, numericals, formulas, and exam-focused concepts.

1. Introduction

Newton's Laws of Motion are three fundamental laws formulated by Sir Isaac Newton that describe the relationship between the motion of an object and the forces acting on it. They form the foundation of classical mechanics.

The three laws are: (1) Newton's First Law — Law of Inertia, (2) Newton's Second Law — Law of Acceleration, and (3) Newton's Third Law — Law of Action and Reaction.

2. Basic Concepts

2.1 Force

A force is an external influence that can change or tend to change the state of motion of an object. It can start or stop an object, change its speed or direction, or deform it.

SI unit: **Newton (N)**

$$1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$$

2.2 Mass

Mass is the quantity of matter in an object. Its SI unit is kilogram (kg). Mass is a scalar quantity and is also a measure of an object's inertia.

2.3 Momentum

Momentum is the quantity of motion possessed by an object.

$$p = mv$$

where p is momentum, m is mass, and v is velocity. Momentum is a vector quantity; SI unit: $\text{kg} \cdot \text{m/s}$.

2.4 Acceleration

Acceleration is the rate of change of velocity with respect to time.

$$a = (v - u) / t$$

3. Newton's First Law of Motion

Statement: An object remains at rest or continues to move with uniform velocity in a straight line unless acted upon by an external unbalanced force.

It is also called the **Law of Inertia**.

$$\Sigma F = 0 \Rightarrow a = 0$$

Therefore, the object either remains at rest or moves with constant velocity.

Types of Inertia

Inertia of rest: A stationary object tends to remain stationary. Example: passengers fall backward when a stationary bus suddenly starts.

Inertia of motion: A moving object tends to continue moving. Example: passengers fall forward when a moving bus suddenly stops.

Inertia of direction: An object tends to continue moving in the same direction. Example: passengers move outward when a car takes a sharp turn.

Applications

Seat belts, dusting carpets, shaking tree branches, and sudden starting or stopping of vehicles are common applications of the first law.

4. Newton's Second Law of Motion

Statement: The rate of change of momentum of an object is directly proportional to the applied external force and occurs in the direction of the applied force.

$$F \propto dp/dt$$

$$F = dp/dt$$

For constant mass, $p = mv$, hence $F = m(dv/dt) = ma$.

$$\mathbf{F = ma}$$

Meaning of $F = ma$

For constant mass, acceleration is directly proportional to net force and inversely related to mass. The acceleration is in the direction of the net force.

Net Force

Newton's second law uses the **net external force**, not necessarily one individual force.

$$F_{\text{net}} = F_1 + F_2 + F_3 + \dots$$

5. Newton's Second Law — Numerical Examples

Example 1: A force of 20 N acts on a 5 kg object. Find acceleration.

$$a = F/m = 20/5 = \mathbf{4 \text{ m/s}^2}$$

Example 2: A 10 kg object accelerates at 3 m/s². Find force.

$$F = ma = 10 \times 3 = \mathbf{30 \text{ N}}$$

Example 3: A force of 50 N produces acceleration of 5 m/s². Find mass.

$$m = F/a = 50/5 = \mathbf{10 \text{ kg}}$$

Example 4: Forces of 20 N right and 8 N left act on a 4 kg object. Net force = 12 N right, so $a = 12/4 = \mathbf{3 \text{ m/s}^2}$ right.

6. Newton's Third Law of Motion

Statement: For every action, there is an equal and opposite reaction.

$$F_{AB} = -F_{BA}$$

Action and reaction forces have equal magnitude, opposite direction, act simultaneously, act on different objects, and are forces of the same type.

Important: Action and reaction do not cancel each other because they act on different objects.

Examples

Walking: The foot pushes the ground backward; the ground pushes the foot forward.

Swimming: The swimmer pushes water backward; water pushes the swimmer forward.

Rocket propulsion: The rocket pushes gases downward; the gases exert an upward force on the rocket.

Gun recoil: The bullet is accelerated forward while the gun experiences a backward reaction force.

7. Free-Body Diagrams and Common Forces

A Free-Body Diagram (FBD) represents all external forces acting on a selected object.

Common forces include weight, normal reaction, friction, tension, applied force, and air resistance.

Weight

$$W = mg$$

Near Earth's surface, $g \approx 9.8 \text{ m/s}^2$.

Normal Force

The normal force is a contact force perpendicular to a surface. For an object on a horizontal surface with no vertical acceleration or additional vertical forces, $N = mg$.

Friction

$$f \leq \mu_s N \text{ and } f = \mu_k N$$

Maximum static friction is $f_{s,\text{max}} = \mu_s N$. Usually $\mu_s > \mu_k$.

Tension

Tension is the pulling force transmitted through a stretched string, rope, or cable. For an ideal massless string and frictionless pulley, tension is often treated as uniform.

8. Elevator Problems

For a person of mass m in an elevator, the major forces are normal force N upward and weight mg downward.

Elevator accelerating upward:

$$N - mg = ma \Rightarrow N = m(g + a)$$

The apparent weight increases.

Elevator accelerating downward:

$$mg - N = ma \Rightarrow N = m(g - a)$$

The apparent weight decreases.

Constant velocity: $a = 0$, so $N = mg$.

9. Circular Motion

For circular motion, acceleration is directed toward the center of the circle.

$$a_c = v^2/r$$

$$F_c = ma_c = mv^2/r$$

Centripetal force is not a new type of force. It is the net inward force responsible for circular motion and may be supplied by tension, gravity, friction, normal force, or another actual force.

10. Impulse and Momentum

Newton's second law can be written in momentum form:

$$F = dp/dt$$

Impulse is force multiplied by the time interval for which the force acts.

$$J = F\Delta t = \Delta p$$

Airbags and crumple zones increase the time over which momentum changes during a collision, reducing the average force.

11. Conservation of Momentum

If no external net force acts on a system, total momentum remains constant.

$$p_{\text{initial}} = p_{\text{final}}$$

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

Applications include collisions, explosions, recoil, and rocket propulsion.

12. Comparison of Newton's Three Laws

First Law: Describes motion when net force is zero — concept of inertia.

Second Law: Describes how net force changes motion — $F = ma$.

Third Law: Describes interaction between two bodies — equal and opposite forces.

13. Problem-Solving Method

1. Identify the object/system. 2. Draw the free-body diagram. 3. Choose coordinate axes. 4. Resolve angled forces into components if needed. 5. Apply $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$. 6. Solve the equations. 7. Check units, magnitude, and direction.

14. Important Conceptual Questions

Can an object move with zero acceleration? Yes. An object moving with constant velocity in a straight line has $a = 0$ even though $v \neq 0$.

Can an object have zero velocity but non-zero acceleration? Yes. At the highest point of a vertically thrown ball, velocity is momentarily zero but acceleration is g downward.

Why don't action and reaction cancel? They act on different objects.

Which has greater inertia: 5 kg or 50 kg? The 50 kg object.

Does an object need force to keep moving? Not if it is already moving with constant velocity and the net force is zero.

15. Formula Sheet

$$\text{Force: } F = ma$$

$$\text{Momentum: } p = mv$$

$$\text{Second law: } F = dp/dt$$

$$\text{Weight: } W = mg$$

$$\text{Acceleration: } a = (v - u)/t$$

$$\text{Kinetic friction: } f_k = \mu_k N$$

$$\text{Maximum static friction: } f_{s,\max} = \mu_s N$$

$$\text{Centripetal acceleration: } a_c = v^2/r$$

$$\text{Centripetal force: } F_c = mv^2/r$$

$$\text{Impulse: } J = F\Delta t = \Delta p$$

$$\text{Conservation of momentum: } p_{\text{before}} = p_{\text{after}}$$

16. Practice Questions

1. Define force and give its SI unit.
2. State Newton's three laws of motion.
3. What is inertia? Explain its three types.
4. Derive $F = ma$ from the momentum form of Newton's second law.
5. A force of 50 N acts on a 10 kg body. Find its acceleration.
6. A 20 kg object accelerates at 2.5 m/s^2 . Find the net force.
7. Forces of 30 N right and 10 N left act on a 5 kg body. Find acceleration.
8. A 2 kg object changes velocity from 5 m/s to 15 m/s in 2 s. Find the force.
9. Explain why passengers fall forward when a moving bus suddenly stops.
10. Explain rocket propulsion using Newton's third law.
11. Why do action and reaction forces not cancel each other?
12. Explain the role of seat belts using Newton's first law.